

Math 220: Mathematical Reasoning and Proof
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All Solutions

Last updated: August 24, 2026



1 (due September 4): Introduction to course. First proof. Logic

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Solutions 1: Introduction to course. First proof. Logic

Solutions to appear after the assignment is due.

1. Using the definitions of “divides” and “even”, explain why 0 is even. Explain why 7 divides 0, but 0 does not divide 7.
2. Suppose that n is an even integer, and let m be any integer. Prove that nm is even.
3. Suppose that n and m are odd integers. Prove that $n + m$ is an even integer. (Hint: an integer n is odd if and only if there exists an integer k such that $n = 2k + 1$.)
4. After reading the syllabus, write out the statement: “I read the syllabus and am aware of this course’s policies”.